

Week 0 - Intro to Machine Learning FoundationsImp. Topics from Mathematics:-

- 1) Differentiability and Derivative
- 2) Critical Points, Local Maxima and Minima
- 3) Vectors, Matrix, System of Linear equation determinants
- 4) Solution of a system and Linear Equations the Gauss-Jordan elimination method
- 5) Vector Spaces, Linear dependence and Independence Basics
- 6) Null space of a Matrix, rank/dimension
- 7) Equivalence and Similarity of Matrices
- 8) Orthogonality and Linear Independence, Projection using inner products.
- 9) Directional Derivatives in terms of the gradient,
- 10) Partial Derivatives.
- 11) Direction of Steepest Ascent/Descent

Imp. Topics from Statistics:-

- 1) Basics of Probability
- 2) Conditional Probability
- 3) Bayes Rule
- 4) Discrete Random Variables
- 5) P.M.F
- 6) Expectation and Variance
- 7) Continuous Random Variables
- 8) Statistics from Samples and Limit Theorems
- 9) Parameter Estimation
- 10) Bayesian Estimation and Hypothesis Testing
- 11) Hypothesis Testing

Week 1 (Lecture 1)1) What is Machine Learning?

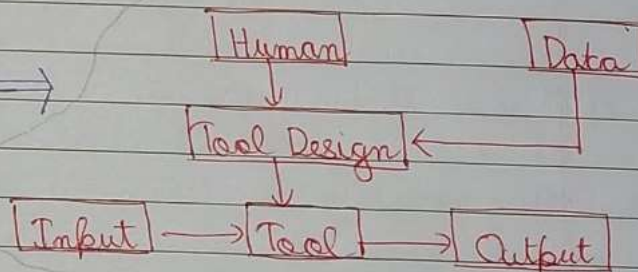
Machine Learning (ML) is the study of computer algorithms that improve automatically through experience and by the use of data.

eg of ML tasks are whether prediction, face detection.

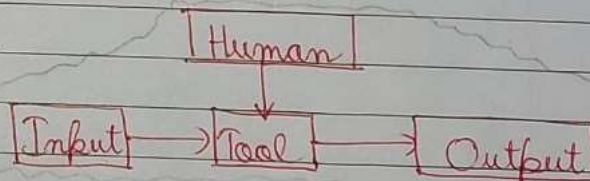
Task → ~~Pro~~ Converting input into an output

Task Hierarchy:-

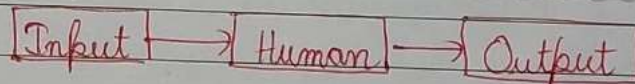
1) Machine Learning ⇒



2) (Programming) + (Tool Usage) ⇒



3) Manual Labour ⇒

Why and when Machine Learning?1. Programming / Human Labour Fails

1. Scale / Speed / Cost of Human Labour
- 2) Inability to express rules using language
- 3) Don't know the exact rules transferring input to output.

2) Machine Learning can succeed

- 1) Have lots of example data
- 2) Have some structural ideas on the rules.



## W2-L2 Data Model and ML Tasks:-

Q-1 What is data?

A collection of Vector (In context of ML)

|    |   |    |     |     |         |
|----|---|----|-----|-----|---------|
| eg | 3 | 9  | 1.9 | 5.0 | House 1 |
|    | 2 | 7  | 2.1 | 3.2 | House 2 |
|    | 4 | 12 | 2.8 | 6.6 | House 3 |
|    | 5 | 16 | 0.9 | 9.8 | " 4     |
|    | 5 | 15 | 3.1 | 8.5 | " 5     |
|    | 4 | 11 | 1.6 | 6.9 | " 6     |

Metadata is information on the data

eg:- (rooms, Area in 100 sq.-ft., Distance from metro, Price in lakhs)

Q-2 What is a Model?

A model is a mathematical simplification of reality

for eg. Ideal Gas Model, Inverse sq. law for gravitational constant  
Moseley's Law for Semiconductors, Cobb-Douglas Model

Types of Model in ML:-

1) Predictive Model

a) Regression Model

b) Classification Model

2) Probabilistic Model

Regression Model:- Model of the price of a house based on its area and distance to metro.

eg. Good Model:  $\text{Price} = 0.5 * \text{Area} - \text{Distance}$   
(Regression Model) → Predictive in nature with real valued output.

Predictive Model:-

Model a house whether a house is closer than 2 km to a metro based on price and area.

Eg. of Good Model

$$\text{Ans} = \begin{cases} \text{Close} & \text{if } 2 * \text{Rooms} - \text{Price} < 1 \\ \text{Far} & \text{otherwise} \end{cases}$$

Probabilistic Models:- (How likely) (Scores different configurations)

Q-1) What's the probability that a randomly chosen person is in latitude-longitude: (25N, 30E)?

Q-2) What's the probability that the tweet was done by Chaplin

Learning Algorithms:-

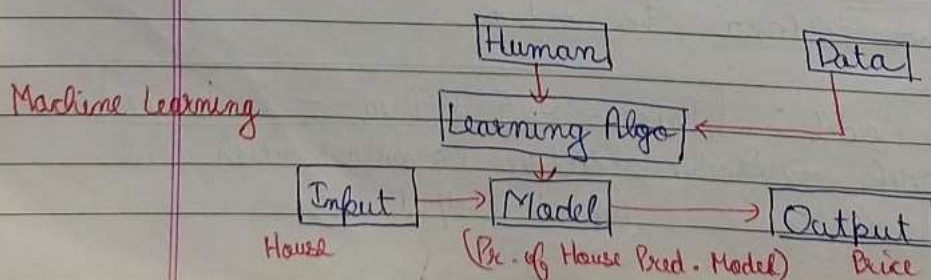
Learning Algorithms: Data  $\rightarrow$  Models

Choose from a collection of models, with same structure but different parameters.

eg.  $\text{Price} = a * (\text{area}) + b * (\# \text{rooms}) + c * (\text{dist. from metro})$

Parameters:  $a, b, c$

Use data to get the "best" Parameters





AQ 1.2:-

1) Best suited for regression?

- Pr. of house based on its area & dist. from metro
- Whether or not

2) In supervised learning, the training labels are available

3) Supervised learning is essentially

4

4) Regression is used when output variable is?  
Continuous.1.1.3 Supervised Learning: Regression

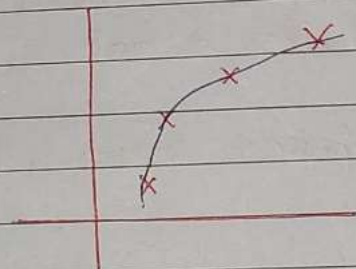
Main paradigm (pattern) behind ML

Notations:-  $R$ : Real No.,  $R_+$ : Positive reals•  $R^d$ :  $d$ -dimensional vectors of real. eg.  $R^3 = \begin{bmatrix} 1.2 \\ 2.3 \\ 1.1 \end{bmatrix}$ •  $x$ : vector,  $x_j$ :  $j^{\text{th}}$  co-ordinate,  $\|x\|$ : length of vector  $x$ .  
(norm)•  $x^1, x^2, \dots, x^n$ : Collection of  $n$  vectors•  $x_j^i$ :  $j^{\text{th}}$  coordinate of  $i^{\text{th}}$  vector eg.  $x_2^3 \rightarrow$  (2nd elem. of 3rd vector)•  $(x_1)^2$ : Square of the first co-ordinate of vector  $x$ .•  $1(\text{? is even}) = 1$ ,  $1(\text{? is odd}) = 0$

## Supervised Learning:-

- It is curve-fitting at its core

- Given  $\{(x^1, y^1), (x^2, y^2), \dots, (x^n, y^n)\}$



- Find a model  $f$  such that  $f(x^i)$  is 'close' to  $y^i$ .  
( $f: \mathbb{R}^d \rightarrow \mathbb{R}$ )

## eg of Regression:-

eg. Predict house price from room, area, distance

- Training data:  $\{(x^1, y^1), (x^2, y^2), \dots, (x^n, y^n)\}$

- $x^i \in \mathbb{R}^d$ ,  $y^i \in \mathbb{R}$

- Algorithm outputs a model  $f: \mathbb{R}^d \rightarrow \mathbb{R}$

- Loss =  $\frac{1}{n} \sum_{i=1}^n (f(x^i) - y^i)^2$

- $f(x) = w^T x + b = \sum_{j=1}^d w_j x_j + b$

$$f(x) = w_1 x_1 + w_2 x_2 + \dots + w_d x_d$$

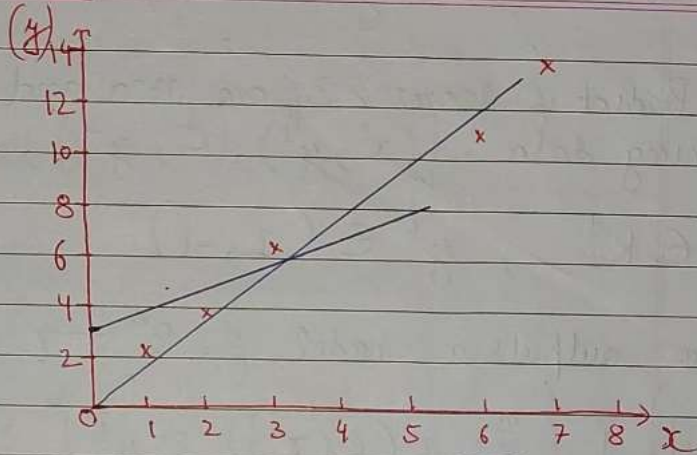
$$= w_1(\text{rooms}) + w_2(\text{area}) + w_3(\text{dist.}) + P$$

Learning algo will find best  $w_1, w_2, w_3, P$  for training data.



Regression Illustration :- $d=1$ 

| $x$ | $y$  |
|-----|------|
| [1] | 2.1  |
| [2] | 3.9  |
| [3] | 6.2  |
| [6] | 11.5 |
| [7] | 13.9 |



Learning algo. will take data and form  $f$  which will have less loss.

Q-1 Suppose

$$f(x) = 2x_1$$

$$g(x) = x_1 + 3$$

Find best suited model among these two.

$$\text{Loss}[f] = \frac{1}{5} ((2-2.1)^2 + (4-3.9)^2 + \dots)$$

$$\text{Loss}[g] = \frac{1}{5} [ \dots ]$$

Illustration 2 :-

| Rooms | Area | Dist. | Price | $f = 2 \times \text{Rooms} - 0.5 \text{ dist.}$ | $g = \text{rooms} + 2 \times \text{dist.}$ |
|-------|------|-------|-------|-------------------------------------------------|--------------------------------------------|
| 3     | 9    | 1.9   | 5.0   |                                                 |                                            |
| 2     | 7    | 2.1   | 3.2   | 5                                               | 6.8                                        |
| 4     | 12   | 2.8   | 6.6   | 3                                               | 6.2                                        |
| 5     | 16   | 0.9   | 9.8   | 6.6                                             | 9.6                                        |
| 5     | 15   | 3.1   | 8.5   | 9.5                                             | 6.8                                        |
| 4     | 11   | 1.6   | 6.9   | 8.5                                             | 11.2                                       |
|       |      |       |       | 7.2                                             | 7.2                                        |

$$\text{Loss}[f] = (5-5.0)^2 + (3-3.2)^2$$

$$\text{Loss}[g] =$$

$\Rightarrow f$  is better.

## Supervised learning : Classification :-

- eg. Predict if rooms > 3 from area and price
- Training data :  $\{(x^1, y^1), (x^2, y^2), \dots, (x^n, y^n)\}$
- $x^i \in \mathbb{R}^d$ ,  $y^i \in \{+1, -1\}$
- Algorithm outputs a model  $f: \mathbb{R}^d \rightarrow \{+1, -1\}$
- $\text{Loss}[f] = \frac{1}{n} \sum_{i=1}^n \mathbb{1}(f(x^i) \neq y^i)$
- $f(x) = \text{Sign}(w^T x + b)$  (Linear Separator)

### Classification Illustration :-

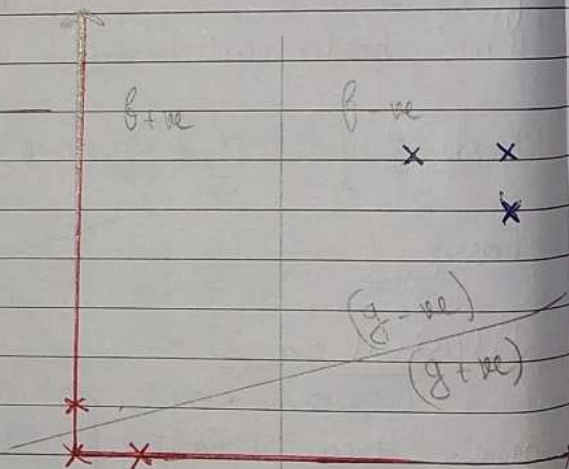
| $x$      | $y$ | $g$ | $f$ | $g$ |
|----------|-----|-----|-----|-----|
| $[0, 0]$ | 1   | 1   | 1   | 1   |
| $[1, 0]$ | 1   | 1   | 1   | 1   |
| $[0, 1]$ | 1   | 1   | 1   | -1  |
| $[4, 4]$ | -1  | -1  | -1  | -1  |
| $[3, 4]$ | -1  | -1  | -1  | -1  |
| $[4, 3]$ | -1  | -1  | -1  | -1  |

$$f(x) = \text{Sign}(2 - x_1)$$

$$g(x) = \text{Sign}(x_1 - 2x_2)$$

$$\text{Loss}[f] = \frac{1}{6}(0) \quad , \quad \text{Loss}[g] = \frac{1}{6}(1) = 0$$

(No. of times you get sign wrong)





Classification Illustration 2

| Area | Price | Rooms | f  | g  | h  |
|------|-------|-------|----|----|----|
| 9    | 5.0   | -1    | -1 | -1 | -1 |
| 7    | 3.1   | -1    | -1 | -1 | -1 |
| 12   | 6.9   | 1     | -1 | -1 | -1 |
| 16   | 9.7   | 1     | 1  | 1  | -1 |
| 15   | 8.5   | 1     | 1  | 1  | 1  |
| 11   | 7.1   | 1     | 1  | 1  | -1 |
|      |       |       | 1  | 1  | -1 |

$$\begin{aligned} f(x) &= \text{Sign}(\text{area} - 10) \\ g(x) &= \text{Sign}(\text{price} - 6) \\ h(x) &= \text{Sign}(\text{price} - 9) \end{aligned}$$

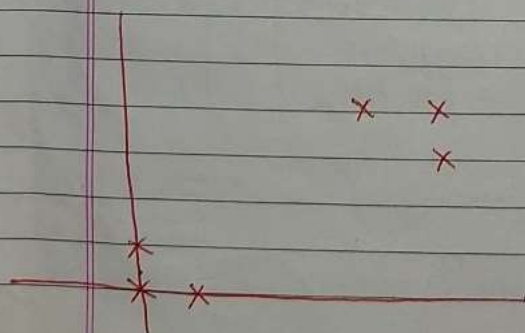
$$\text{Loss}[f] = 0, \text{Loss}[g] = 0$$

$$\text{Loss}[h] = \frac{1}{2}$$

(For this training data any among them can be chosen)

Evaluating Learned Models : Test data

- Learning algo uses training data  $(x^1, y^1), \dots, (x^n, y^n)$  to get the model  $f$ .
- But evaluating the learned model must not be done on the training data itself.
- Use test data that is not in the training data for model evaluation.



$$f(x) = \begin{cases} +1 & \text{if } x = \begin{bmatrix} 0 \\ 0 \end{bmatrix} \\ +1 & \text{if } x = \begin{bmatrix} 1 \\ 0 \end{bmatrix} \\ +1 & \text{if } x = \begin{bmatrix} 0 \\ 1 \end{bmatrix} \\ -1 & \text{otherwise} \end{cases}$$

### Model Selection : Validation Data

- Learning algo just finds the 'best' model in the collection of models given by humans.
- How to find right collection of models?
- This is called model selection, and it is done by using another subset of data called validation data, that is distinct from train and test data.

$$\text{Price} = a \times \frac{\text{area}}{\text{rooms}} + b (\text{distance})^2 + c \left( \frac{\text{distance}}{\text{area}} \right)$$

#### AQ 1.9

- 1) Classification models are Predictive
- 2) Which of foll. loss f<sup>n</sup> can be applied to classif. model?  

$$\frac{1}{n} \sum_{i=1}^n (f(x^i) - y^i)^2$$
- 3) We evaluate our model on which of foll. data?  
Test data
- 4) Model Selection is done using which of the foll. data?  
Validation Data



## Unsupervised Learning : Dimensionality Reduction

- Unsupervised learning is 'understanding data'
- Data :  $\{x^1, x^2, \dots, x^n\}$
- $x^i \in \mathbb{R}^d$
- Build models that compress, explain and group data.

### Dimensionality Reduction -

eg. Represent a million gene expressions<sup>levels</sup> of a million people, using just 100 no. per person.

Dimensionality red<sup>n</sup> : compression & simplification.

- Data :  $\{x^1, x^2, \dots, x^n\}$  ( $d' < d$ )
- $x^i \in \mathbb{R}^d$
- Encoder  $f: \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$
- Decoder  $g: \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$
- Goal :  $g(f(x^i)) \approx x^i$
- Loss =  $\frac{1}{n} \sum_{i=1}^n \|g(f(x^i)) - x^i\|^2$

### Dimensionality Red<sup>n</sup> Illustration

$d=2, d'=1$

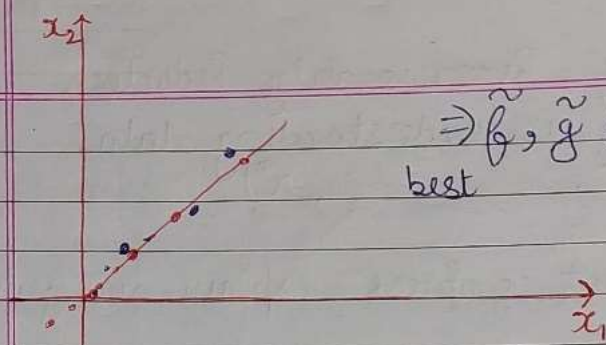
| $x$        | $f$  | $g$            | $x$        | $\tilde{f}$ | $\tilde{g}$  |
|------------|------|----------------|------------|-------------|--------------|
| $[1, 0.8]$ | 0.2  | $[0.2, 0.2]$   | $[1, 0.8]$ | 0.9         | $[0.9, 0.9]$ |
| $[2, 2.2]$ | -0.2 | $[-0.2, -0.2]$ | $[2, 2.2]$ | 2.1         | $[2.1, 2.1]$ |
| $[3, 3.2]$ | -0.2 | $[-0.2, -0.2]$ | $[3, 3.2]$ | 3.1         | $[3.1, 3.1]$ |
| $[4, 3.8]$ | 0.2  | $[0.2, 0.2]$   | $[4, 3.8]$ | 3.9         | $[3.9, 3.9]$ |

$$f(x) = x_1 - x_2$$

$$g(u) = [u, u]$$

$$\tilde{f}(x) = \frac{x_1 + x_2}{2}$$

$$\tilde{g}(u) = [u, u]$$



$\Rightarrow \tilde{f}, \tilde{g}$  performs a best

We're asking dimensionality red<sup>n</sup> algo<sup>cm</sup> to choose best among these two.

1.5 1) Unsupervised learning is essentially:-  
Understanding Data

2) Let  $f: \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$  be an encoder  $f^n$  for dimensionality red<sup>n</sup>, then what is the typical relation b/w  $d$  and  $d'$ ?

$$d > d'$$

3) Let  $f: \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$  be a decoder  $f^n$ , then all is?

$$d' > d \quad \text{or} \quad d' < d$$

4) for data  $\{x^1, x^2, x^3, \dots, x^n\}$  where  $x^i \in \mathbb{R}^d$  and  $f: \mathbb{R}^d \rightarrow \mathbb{R}^{d'}$ ,  $g: \mathbb{R}^{d'} \rightarrow \mathbb{R}^d$ , which of the foll. is correct loss  $f^n, g^n$  for a dimensionality red<sup>n</sup> algo?

$$\frac{1}{n} \sum_{i=1}^n \|f(g(x^i)) - x^i\|^2$$

$$\frac{1}{n} \sum_{i=1}^n \|g(f(x^i)) - x^i\|^2$$



## Unsupervised learning : Density estimation

Its output is probabilistic Model (scores tweets)

eg. Assuming tweets from an account are independently generated randomly. Create a robot account that generates more such tweets.

Supposing a tweet to be 128 letter array, then

$$p(x) \text{ such that } x \in [26]^{128}$$

here to generate such sentence randomly, we need to be able to assign a probability score to every possible 128 character sentence, giving high scores to those that are likely to be from the original source.

A density estimation model takes in several samples from a random source, and outputs a model that assigns a probability score to every possible instance.

eg.  $P(\text{"Apple is bad"})$

- Data :  $\{x^1, x^2, \dots, x^n\}$

- $x^i \in \mathbb{R}^d$

- Probability mapping  $P: \mathbb{R}^d \rightarrow \mathbb{R}_+$  that 'sums' to 1.

- Goal :  $P(x)$  is large if  $x \in \text{Data}$ , and low otherwise.

- Loss =  $\frac{1}{n} \sum_{i=1}^n -\log(P(x^i))$  : Neg. log likelihood

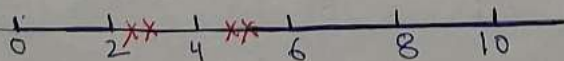
Density Estimation Illustration 1 :-  $d=1, n=4$

$$x^1 = [2.3]$$

$$x^2 = [2.7]$$

$$x^3 = [4.6]$$

$$x^4 = [4.9]$$



$$P(X) = \begin{cases} \frac{1}{10} & \text{if } x \in [0, 10] \\ 0 & \text{otherwise} \end{cases}$$

Page No. 16

Date / /

$$P^2(X) = \begin{cases} \frac{1}{5} & \text{if } x \in [0, 5] \\ 0 & \text{otherwise} \end{cases}$$

$$P^3(X) = \begin{cases} \frac{1}{5} & \text{if } x \in [3, 8] \\ 0 & \text{otherwise} \end{cases}$$

|     | $P^1$          | $P^2$         | $P^3$         |
|-----|----------------|---------------|---------------|
| 2.3 | $\frac{1}{10}$ | $\frac{1}{5}$ | 0             |
| 2.6 | $\frac{1}{10}$ | $\frac{1}{5}$ | 0             |
| 4.6 | $\frac{1}{10}$ | $\frac{1}{5}$ | $\frac{1}{5}$ |
| 4.9 | $\frac{1}{10}$ | $\frac{1}{5}$ | $\frac{1}{5}$ |

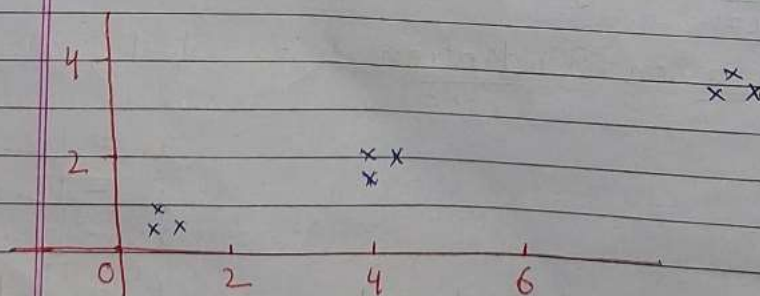
$$\text{loss}[P^1] = -\log\left(\frac{1}{10}\right) - \log\left(\frac{1}{10}\right) - \log\left(\frac{1}{10}\right) - \log\left(\frac{1}{10}\right)$$

$$\text{loss}[P^2] = -\log\left(\frac{1}{5}\right) - \log\left(\frac{1}{5}\right) - \log\left(\frac{1}{5}\right) - \log\left(\frac{1}{5}\right)$$

$$\text{loss}[P^3] = -\log(0)$$

(Since loss of  $P^2$  is smaller than loss of  $P^1$  it is better.)

Q-2) You have 9 data points -





$P^1 = \text{Gaussian Mixture Model}$   
 $= \text{GMM} \left( \begin{bmatrix} 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 4 \\ 1 \end{bmatrix}, \begin{bmatrix} 7 \\ 3 \end{bmatrix} \right)$

Page No. 17

Date / /

$$P^2 = \text{G.M.M} \left( \begin{bmatrix} 5 \\ 5 \end{bmatrix}, \begin{bmatrix} 8 \\ 9 \end{bmatrix}, \begin{bmatrix} -1 \\ -2 \end{bmatrix} \right)$$

(From graph as well as -ve log likelihood  $P^1$  has less loss)

Q1.6 1) Input data from an unsupervised learning algo is of the form (Note:  $X^i$ 's and  $Y^i$ 's are vectors,  $y^i$  are scalars):

$$\{X^1, X^2, \dots, X^n\}$$

2) Which gives a probabilistic model?  
 Density Estimation

3) Correct loss for density estimation?  
 $\frac{1}{n} \sum_{i=1}^n -\log(P(X^i))$

4) During Parameter estimation for a density estimation method such as GMM, which of foll. is minimized?

Neg. log likelihood

Tutorial Ques:- Look at the data table below and decide whether a ML approach is required to solve this problem.

| Data             | Label |
|------------------|-------|
| $S1 = [3, 5, 4]$ | 0     |
| $S2 = [3, 4, 5]$ | 1     |
| $S3 = [4, 2, 1]$ | 0     |
| $S4 = [6, 7, 8]$ | 1     |
| $S5 = [1, 2, 3]$ | 1     |
| $S6 = [1, 1, 1]$ | 1     |
| $S7 = [5, 2, 0]$ | 0     |

Q is  $\Rightarrow [4, 5, 3] \rightarrow ?$

|     |    |            |                        |
|-----|----|------------|------------------------|
| Yes | No | Stowe More | Not possible to decide |
|-----|----|------------|------------------------|

Secret Revealed :-

Consider a problem of identifying whether an integer sequence of length  $N=3$ , say,  $S=[3, 5, 4]$  is sorted in ascending order or not. Does this problem absolutely require machine learning approach to solve? Justify

Justification :-

- 1) We know the rule and can transform it into code
- 2) Sol<sup>n</sup> is also scalable

Let us pose it as ML problem, then think about the data collection (generation) process.

| Data              | Label |
|-------------------|-------|
| $S_1 = [3, 5, 4]$ | 0     |
| $S_2 = [3, 4, 5]$ | 1     |
| $S_3 = [2, 1, 4]$ | 0     |
| $S_4 = [6, 7, 8]$ | 1     |
| $S_5 = [1, 2, 3]$ | 1     |
| $S_6 = [1, 1, 1]$ | 1     |
| $S_7 = [1, 2, 0]$ | 0     |

Metadata associated with dataset?

- No. of dataset points: 1000
- Element type: Non-negative integers
- Licence: CC

Since we have data as well as label we'll use supervised learning approach

↓  
Since labels can only take values 0 or 1

⇒ It is classification



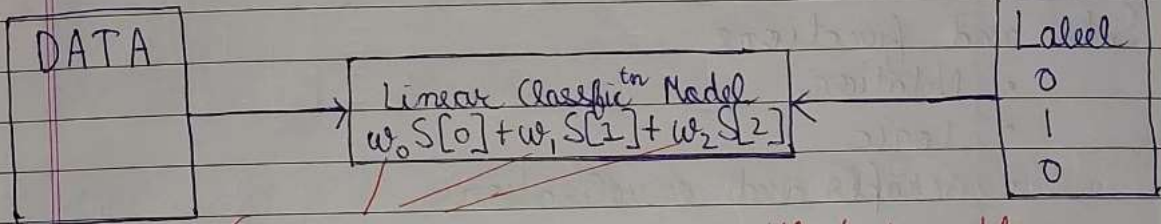
Assumption:- You have only the "data and corresponding labels", no other information.

Page No. 19

Date / /

Model: 1> Hand Crafted Model

2> ML Model



(Learning Model will come with best models among these  $\{w_0, w_1, w_2\}$  i.e. (parameter or weight) of model. The best values for the parameters will be learned from the data)

# Notes By Hardik Sharma

If any part is blurred or not understandable  
message me on WhatsApp (9877100914)

Tip: Don't directly jump onto the notes firstly  
watch the whole video and then read them.